

1. RAG (Retrieval-Augmented Generation)

RAG is an architectural pattern that gives a Large Language Model (LLM) access to external data sources to answer questions accurately.

The Problem: LLMs are frozen in time and do not know about your private data, company documents, or real-time web events. They also tend to hallucinate (make up facts).

- **The Solution:** Instead of sending a user's question directly to the LLM, a RAG system first searches a database (like ChromaDB) for documents related to that question. It then glues the question and the retrieved text chunks together, passing them to the LLM as a single prompt.
- **The Result:** The LLM acts as an "open-book" reader. It answers using only the provided facts, ensuring up-to-date and highly verified responses without needing expensive model retraining.